

PHYS114 Reference

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1 Data Files, Software, and Figures

1.1 Data Files

- **Binary:**
 - Advantages: compact file size.
 - Disadvantages: need encoders (otherwise difficult to read).
 - Microsoft 365 files (.xlsx, .docx, etc.), which are proprietary but have been open-sourced for interoperability with, for example, Apple and Google's suites.
 - **.fits:** Well-known, well-defined, and used by the astronomy community.
 - **.cdf:** Common Data Format, standardized encoding.
 - **.sav:** IDL's proprietary format.
- **ASCII:**
 - Advantages: built-in encoding table and, thus, easy to read.
 - Disadvantages: large file size.
 - .txt, .csv, etc.

1.2 Software

- A wide array of scientific software exists such as MATLAB and IDL.
- However, these tend to require expensive licenses.
- In contrast, Python is free and open-source.
- However, Python comes with no warranty, meaning that if something goes wrong, the user is responsible.

1.3 Figures

A professional-quality figure

- is grayscale, relying on various marker sizes and shapes (e.g. circle, diamond, etc.) as well as line styles (solid, dashed, etc.) to differentiate different series.
- contains a title, axes labels with units, and a legend as necessary.
- uses space on the plot effectively.
- has proper axis scaling.
- is high resolution (usually 600 DPI or more).

2 Introduction to Research

Some information is specific to the New Jersey Institute of Technology (NJIT).

- A typical external federal grant is \$130,000 for 3 years.
- Professors must bring in grants to be tenured.
- Professors also have a "2 + 2" teaching requirement: 2 courses in the fall, and 2 courses in the spring.
- If a professor supports a research assistant (RA), covering their **stipend**, **tuition**, and **fringe**, they go down to 1 + 2.
- Supporting another RA reduces this to 1 + 1.
- In contrast, teaching assistants (TAs) are paid for by the university, not the professor.
- A researcher or RA typically cannot have another job if they are already being funded by a grant for maximal dedication.
- Usually, grant money goes to the university, not the professor directly.
- PhDs in the United States usually take 6 years because new knowledge must be created to attain it.
- A typical physics graduate student coursework:
 - Fall: Electricity and Magnetism I, Classical Mechanics, Mathematical Methods
 - Spring: Electricity and Magnetism II, Quantum Mechanics, Statistical Mechanics / Thermodynamics

- Qualifying exams in the summer (June for NJIT) determine whether a student remains in the graduate program.
- Qualifying exams are difficult (typically 50% pass rate for NJIT).
- There is one last redemption exam in January for those who fail.
- In the 1980s, program officers would evaluate grant proposals, but from 1990s onward, panels of qualified individuals perform this task.

2.1 Federal Funding Agencies

- Top agencies that fund grants:
 - National Science Foundation (NSF)
 - Department of Energy (DOE)
 - National Aeronautics and Space Administration (NASA)
 - National Institutes of Health (NIH)
 - Department of Defense (DoD)
- All except NSF require **closure**, which means meaningful results must be found, and the research question must be answered.
- Some agencies require an American citizenship for all researchers.

2.2 Grant Format

- A small part of the grant ($\sim 10\%$) is:
 - Project summary (1 page)
 - Project description (15+ pages)
 - Export control, etc.
- But the majority of the grant ($\sim 90\%$) is paperwork

2.3 Fringe and Overhead

- **Fringe** is the cost of employing someone, which includes associated costs such as insurance, retirement, etc.
- **Overhead** is the money used to support the research enterprise of an institution, which includes the cost of utilities, supplies, subscriptions, etc.
- The fringe at NJIT went from 43% to 86% in 5 years.
- The cost of research has skyrocketed due to rising insurance premiums and unionization.

3 Forward Model and Photometer

3.1 Forward Model

A mathematical or quantitative description of the problem you're solving. One can use synthetic data and example numbers to test.

3.2 Photometer

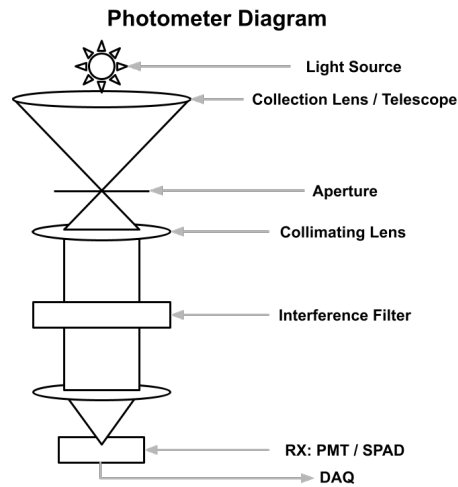


Figure 1: A diagram of a photometer.

- Measures the number of photons from a source.
- IF stands for interference filter.
- RX stands for receiver.
- TX stands for transmitter.
- The receiver can either be a **photomultiplier tube (PMT)** or a **single-photon avalanche diode (SPAD)**. A PMT is a single-element instrument that counts individual photos precisely.
- **Charge-coupled devices (CCDs)** are an array of sensors that take an image of incoming photons. Scientific-grade CCDs are more expensive than PMTs.
- **Photon count formula:**

$$N = N_{\text{signal}} + N_{\text{noise}}$$

where

- N is the total number of photons.
- N_{signal} is the number of photons from the desired source.
- N_{noise} is the number of photons from other (undesirable) sources.

- **Noise photon count formula:**

$$N_{\text{noise}} = N_C \Delta t$$

where

- N_C [counts/s] is the noise count rate.
- Δt [s] is the **integration time**, the time interval at which the detector collects photons.

- **Signal photon count formula:**

$$N_{\text{signal}} = BA\Delta t \Omega T_A \nu_{RX}$$

$$\Omega = \frac{A}{r^2} = 2\pi(1 - \cos \alpha)$$

where

- B [photons/(s · m² · str)] is a constant.
- A [m²] for N_{signal} is the area of the telescope.
- Δt [s] is the integration time.
- Ω [str] is the solid angle.
- T_A is the atmospheric transmission.
- ν_{RX} is the receiver efficiency.
- A for N_{noise} is the area of the detector.
- r is the focal length of the telescope.
- α is the half-angle of the cone of light collected by the telescope, also known as the field of view.

Note: $A\Omega$ is a conversed quality called the **etendue**, and it will be explored later.

4 Distributions and Uncertainty

- For **nonstationary** PDFs, the moments are in terms of time.
- For **wide-sense stationary (WSS)** PDFs, only the **first two moments** are stationary, which represents most real-life phenomena.
- For **stationary** PDFs, all the moments are constant and *not* in terms of time.

- **Moment generating function (continuous):**

$$n^{\text{th}} \text{ moment} = \int_{-\infty}^{\infty} x^n \text{PDF}(x) dx$$

- **Moment generating function (discrete):**

$$n^{\text{th}} \text{ moment} = \sum_{i=1}^{\infty} x_i^n \text{PDF}(x_i)$$

- This class only focuses on the Gaussian, Binomial, and Poisson distributions. The rest are extra information.
- Remember that probabilities multiply. The probability of reproducing an entire dataset is

$$\prod_{i=1}^N \text{PDF}(x_i) = p_1 p_2 \dots p_N$$

- General process of data collection and processing:
 1. Determine PDF.
 2. Create estimators for moments from data.
 3. Determine if estimators are biased or unbiased using **expectation values**.
 4. Determine the chi-squared goodness of fit.
- From strongest to weakest: independent, uncorrelated, and no covariance.
 - **Independent:** no true mathematical representation of independence.
 - **Uncorrelated:** $r \equiv \frac{\sigma_{XY}^2}{\sigma_X \sigma_Y} = 0$
 - **No covariance:** $\sigma_{XY}^2 = 0$

4.1 Gaussian Distribution

Also known as the normal distribution. Represents humans and many natural phenomena.

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right]$$

- μ and σ are parameters.
- 1st moment (mean): μ
- 2nd moment (variance): σ^2

- Estimator for 1st moment:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

- Estimator for 2nd moment:

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

- 3rd moment (skewness)
- 4th moment (kurtosis)
- Uncertainty for 1st moment:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{N}}$$

This signals diminishing returns for increasing N .

- However, the above is assuming WSS. In a case without WSS:

$$\bar{x} = \frac{\sum_{i=1}^N \left(\frac{x_i}{\sigma_i^2} \right)}{\sum_{i=1}^N \left(\frac{1}{\sigma_i^2} \right)}$$

$$\sigma_{\bar{x}}^2 = \frac{1}{\sum_{i=1}^N \frac{1}{\sigma_i^2}}$$

4.2 Binomial Distribution

$$\text{PDF}(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

- Trials must be **independent**.
- For p large and n large, similar to the **Gaussian distribution**.
- For p low and n large, similar to the **Poisson distribution**.
- n and p are parameters.
- 1st moment: np

- 2nd moment: $np(1-p)$
- 3rd moment: $\frac{(1-p)-p}{\sqrt{np(1-p)}}$
- Estimator for 1st moment:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

- Estimator for 2nd moment:

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

4.3 Poisson Distribution

Represents scenarios with counting and waiting for events.

$$\text{PDF}(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

- λ is the parameter.
- 1st moment: λ
- 2nd moment: λ
- 3rd moment: $\frac{1}{\sqrt{\lambda}}$
- Estimator for 1st moment:

$$\bar{x} = \sum_{i=1}^N x_i$$

- Estimator for 2nd moment:

$$s^2 = \sum_{i=1}^N x_i$$

- Percent error:

$$\frac{1}{\sqrt{\bar{x}}} \times 100\%$$

4.4 Laplace Distribution

$$f(x; \mu, b) = \frac{1}{2b} \exp \left[-\frac{x - \mu}{b} \right]$$

- μ and b are parameters.
- μ is a location parameter.
- $b > 0$ is called "diversity."
- 1st moment: μ
- 2nd moment: $2b^2$
- 3rd moment: $\emptyset$

4.5 Gamma Distribution

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

where Γ is the gamma function.

- α and β are parameters.
- α is the shape parameter.
- $\frac{1}{\beta}$ is the scale parameter.
- 1st moment: $\frac{\alpha}{\beta}$
- 2nd moment: $\frac{\alpha}{\beta^2}$
- 3rd moment: $\frac{2}{\sqrt{\alpha}}$

4.6 Assumptions

In this class, we assume that

1. Stationary or WSS PDF: $\sigma_i \approx \sigma$
2. Each measurement is independent and identically distributed (IID).
3. **Method of Maximum Likelihood (MML)**
 - Also known as Maximum Likelihood Method (MLM).
 - Maximum probability of reproducing the data.
 - An alternate method is **Maximum Entropy Method (MEM)** that seeks to add as much disorder as possible to the datapoints, but this class only focuses on MML.
4. An implicit assumption is the zero-mean process. We assume or subtract off the mean to make calculations much easier.

4.7 Expectation Values

$$\langle x_i \rangle = E[x_i] = \mu \quad (\text{typically})$$

Examples:

- We expect the $\bar{x}$ estimator to yield μ : $\langle \bar{x} \rangle = \mu$
- And similarly for s^2 : $\langle s^2 \rangle = \sigma^2$

A rule for expectation values is that you ignore constants:

$$\langle c \rangle = c$$

You can also ignore summations:

$$\left\langle \sum_{i=1}^N x_i \right\rangle = \sum_{i=1}^N \langle x_i \rangle$$

4.8 Uncertainty Propagation

- Uncertainty is represented by σ . For example, the uncertainty in $\bar{x}$ is denoted as $\sigma_{\bar{x}}$.
- Uncertainty can either propagate in a **worst case** or **even-steven** (informally named by our professor) scenario.
- Basics for worst case propagation: Suppose there are height measurements h_1 and h_2 with uncertainties σ_{h_1} and σ_{h_2} . Adding the two heights together also adds their uncertainties: $(h_1 \pm \sigma_{h_1}) + (h_2 \pm \sigma_{h_2}) = (h_1 + h_2) \pm (\sigma_{h_1} + \sigma_{h_2})$. This is the simplest example, but this can get much more complex, especially for operations like multiplication.
- **Even-steven formula** (Taylor series approximation):

$$\sigma_x^2 \approx \sigma_u^2 \left(\frac{\partial x}{\partial u} \right)^2 + \sigma_v^2 \left(\frac{\partial x}{\partial v} \right)^2 + \dots + 2\sigma_{uv}^2 \left(\frac{\partial x}{\partial u} \right) \left(\frac{\partial x}{\partial v} \right) + \dots$$

$$\sigma_{uv}^2 = \frac{1}{N-1} \sum_{i=1}^N (u_i - \bar{u})(v_i - \bar{v})$$

– σ_{uv}^2 is the covariance, which is 0 when both u and v are **independent**.

4.9 Chi-Squared

The following only applies to data that is independent and identically distributed.

4.9.1 Chi-Squared Goodness of Fit

Given a histogram:

$$\chi^2 \equiv \sum_{j=1}^n \left(\frac{h(x_j) - NP(x_j)}{\sigma_j(h)} \right)^2 = \sum_{i=1}^N \frac{(x_i - \bar{x})^2}{\sigma_i^2}$$

where

- n is the number of bins.
- $h(x_j)$ is the value of the j -th bin.
- N is the total number of measurements.
- $P(x_j)$ is the probability of the j -th bin.
- $\sigma_j(h)^2$ is the variance in the j -th bin.

χ^2 is a measure of the goodness of fit of a model to the data. A smaller χ^2 indicates a better fit. Ideal value is 0, but in practice it is around n because the numerator and denominator of the summation are around the same order of magnitude.

4.9.2 Reduced Chi-Squared

$$\chi_\nu^2 = \frac{\chi^2}{\nu}$$
$$\nu = n - n_c$$

- n_c is the number of constraints. Usually 1.
- ν is the number of degrees of freedom. Usually $n - 1$ with no outliers.

4.9.3 Generalized Chi-Squared

$$\chi^2 \equiv \sum_{i=1}^N \frac{(y_i - y(x_i))^2}{\sigma_i^2}$$

where

- y_i is the value of the i -th measurement,
- $y(x_i)$ is the value of the model at the i -th measurement, and
- σ_i^2 is the uncertainty in the i -th measurement.

4.9.4 Chi-Squared Distribution

$$P_{\chi^2 > \chi^2_T}(\chi^2, \nu) = \int_{\chi^2}^{\infty} P_{\chi^2}(\chi^2; \nu) d\chi^2$$
$$P_{\chi^2}(\chi^2, \nu) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} (\chi^2)^{\nu/2-1} e^{-\chi^2/2}$$

where

- ν is the number of degrees of freedom,
- Γ is the gamma function, and
- χ^2 is the chi-squared value. Note that χ^2 is a single value, so do not treat it as *chi* raised to 2.

The purpose of the chi-squared distribution is to determine the probability that the parent model would yield a χ^2 value greater than the one calculated from the data.

On exams, a table will be given to find the probability using the chi-squared distribution.

4.10 Probability of Exceeding a Correlation Coefficient

$$P(r; N) = 2 \int_{|r|}^1 \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi} \Gamma(\frac{\nu}{2})} (1-r^2)^{\frac{\nu-2}{2}} dr$$

where

- Γ is the gamma function,
- r is the correlation coefficient, and
- ν is the number of degrees of freedom. In this case, $\nu = N - 2$.

This is used to determine the probability that the sample appears correlated when it is not.

5 Remote Sensing

Remote sensing is the science of obtaining information about objects or areas from a distance, typically from aircraft or satellites. **Passive** remote sensing "watches" the system, collecting without impacting the system (e.g., photometer). **Active** remote sensing perturbs the system until it reaches steady-state again (e.g., radar). This "nudge" must be small to minimize disruption to the system. Passive remote sensing equipment is preferred as it tends to be less complex than active remote sensing equipment.

5.1 LiDAR

LiDAR, which stands for Light Detection and Ranging, is an active remote sensing method that uses pulses of infrared (or sometimes visible or ultraviolet) light to measure the system. LiDAR offers a higher resolution than radar due to using shorter wavelength of light than radio waves.

5.2 Radar

Radar, which stands for Radio Detection and Ranging, is an active remote sensing method that uses radio waves to measure the system. Radar offers a lower resolution than LiDAR due to using radio waves, which have a longer wavelength.

5.3 Angle of Spread

$$\theta \sim \frac{\lambda}{D}$$
$$\theta = 1.22 \frac{\lambda}{D} \quad \text{for circular apertures}$$

where

- θ is the angle of spread,
- λ is the wavelength of the light,
- D is the diameter of the aperture.

Thus it is clear why radar has a lower resolution than LiDAR, because radio waves have a longer wavelength and thus spread out much more. This leads to a **volume underfilled** scenario, in which the target does not fill the entire volume of the beam.

5.4 Doppler Effect

$$\lambda' = \frac{v - v_s}{f}$$
$$f' = f_0 \left(\frac{v \pm v_o}{v \mp v_s} \right)$$

where

- λ' is the observed shifted wavelength,
- f' is the observed shifted frequency,

- v is the speed of the wave,
- v_o is the speed of the observer,
- v_s is the speed of the source, and
- f'_0 is the emitted frequency.

Sign conventions: In the numerator, use $+$ if the observer is moving towards the source, and $-$ if the observer is moving away from the source. In the denominator, use $-$ if the source is moving towards the observer, and $+$ if the source is moving away from the observer.

5.5 Monostatic and Bistatic Systems

In a **monostatic** system, the transmitter and receiver are at the same location. In a **bistatic** system, the transmitter and receiver are at different locations (usually common for speed detection in rural areas).

5.6 Speed Detection

Speed detection is a common application of remote sensing and can use either radar or LiDAR. Radar speed detection uses the doppler effect to measure the speed of a vehicle, whereas LiDAR speed detection uses the time it takes for the light to return to the sensor to calculate the vehicle's speed.

Radar has a low **signal-to-noise ratio** (SNR) compared to LiDAR at the receiver, allowing a car to detect when its speed is being tracked and jamming the reflected signal (thus, radio detectors have been outlawed in many places). LiDAR has a high SNR (especially when directed at license plates with a special coating), so while a car still detect when its speed is being tracked, it cannot jam the reflected signal (as a result, LiDAR detectors have not been outlawed).

5.7 Stealth

Stealth aircraft conceal themselves to radar detection by using a combination of radar-absorbent materials and clever geometry to deflect radar waves away from the receiver. This is why stealth aircraft are not completely invisible to radar, but they have a very low radar cross-section.

5.8 Jenny Jump

Jenny Jump uses active remote sensing to detect the height that particles are in the atmosphere. The transmitter sends a beam of green light up into the atmosphere 30 times per second. (One pulse would have a low SNR, which is solved by stacking 30 pulses.) The receiver then detects the light that is scattered back to it from the particles in the atmosphere. The height of the particles is then calculated using the time it takes for the light to return to the receiver:

$$z = \frac{c\Delta t}{2}$$

Among other information, this receiver can detect gravity waves (which are a type of buoyancy waves) in the atmosphere. Gravity waves are a cause of clear-air turbulence, which is a hazard for aircraft.

6 Monte Carlo Methods

This class does not focus on Monte Carlo methods, but here is a brief overview. **Monte Carlo methods** are a broad class of computational algorithms that rely on repeated random sampling to estimate possible outcomes of complex, uncertain events. Random number generation is a complex topic, yet it is important for simulations. For example, the simulation of light scattering and weather models predicting a hurricane's path use Monte Carlo methods. Weather models are **ensemble models**, which means that they run the same simulation multiple times with different initial conditions to generate a range of possible outcomes and account for uncertainty. Ensemble models only help for so long until the models start diverging greatly.

7 Fitting a Simple Curves

7.1 Straight Line

Forward model:

$$y = a_0 + b_0x$$

Let σ_i denote the uncertainty in y_i . (The uncertainty in x_i is assumed to be 0.) x_i and y_i are the data points.

Minimize $\Delta y = y_i - y(x_i)$, the difference between the data points and the model. Assume that each y_i is pulled from a stationary Gaussian distribution, and use the maximum likelihood method.

The probability of a particular observation is

$$P_i = \frac{1}{\sigma_i \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{y_i - y(x_i)}{\sigma_i} \right)^2 \right].$$

For a N measurements:

$$\prod_{i=1}^N P_i = \prod_{i=1}^N \left(\frac{1}{\sigma_i \sqrt{2\pi}} \right) \exp \left[-\frac{1}{2} \sum_{i=1}^N \left(\frac{y_i - y(x_i)}{\sigma_i} \right)^2 \right].$$

For the model $y = a_0 + b_0x$:

$$P(a_0, b_0) = \prod_{i=1}^N \left(\frac{1}{\sigma_i \sqrt{2\pi}} \right) \exp \left[-\frac{1}{2} \sum_{i=1}^N \left(\frac{y_i - (a_0 + b_0 x_i)}{\sigma_i} \right)^2 \right]$$

To maximize $P(a_0, b_0)$, we need to minimize the exponent:

$$\chi^2 = \sum_{i=1}^N \left(\frac{y_i - (a_0 + b_0 x_i)}{\sigma_i} \right)^2$$

To minimize χ^2 , we take the partial derivatives with respect to a_0 and b_0 and set them to 0.

$$\frac{\partial \chi^2}{\partial a_0} = 0 \quad \text{and} \quad \frac{\partial \chi^2}{\partial b_0} = 0.$$

The result is:

$$\Delta = \sum \frac{1}{\sigma_i^2} \sum \frac{x_i^2}{\sigma_i^2} - \left(\sum \frac{x_i}{\sigma_i^2} \right)^2$$

$$a_0 = \frac{1}{\Delta} \left(\sum \frac{x_i^2}{\sigma_i^2} \sum \frac{y_i}{\sigma_i^2} - \sum \frac{x_i}{\sigma_i^2} \sum \frac{y_i}{\sigma_i^2} \right)$$

b_0 can be found as well, but the formula is not the focus of this course. The uncertainty on a_0 and b_0 is given by:

$$\sigma_z^2 = \sum \sigma_i^2 \left(\frac{\partial z}{\partial y_i} \right)^2$$

where z can be a_0 or b_0 . Note that the number of constraints n_C is 2 since due to a_0 and b_0 .

7.2 Polynomials

Forward model:

$$y(x_i) = \sum_{k=1}^m a_k f_k(x_i)$$

where $f_k(x_i) = 1, x_i, x_i^2, \dots$ is a polynomial parent function.

Carry out a similar process as fitting a straight line:

$$P = \prod_{i=1}^N \left(\frac{1}{\sigma_i \sqrt{2\pi}} \right) \exp \left[-\frac{1}{2} \sum_{i=1}^N \left(\frac{y_i - \sum_{k=1}^m a_k f_k(x_i)}{\sigma_i} \right)^2 \right]$$

Thus:

$$\frac{\partial \chi^2}{\partial a_\ell} = 0 \quad \text{for } \ell = 1, \dots, m$$

8 Linearization

Linearization is the process of transforming a non-linear function into a linear function:

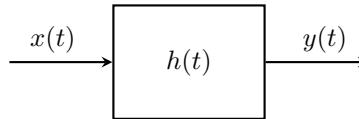
$$\begin{aligned} y &= Ae^{-bx} \\ \ln y &= \ln(Ae^{-bx}) \\ \ln y &= \ln A - bx \end{aligned}$$

9 System Theory

System theory is the study of how individual components of a system interact with each other to produce a collective behavior. A **linear time-invariant (LTI)** system is a system that satisfies the following properties:

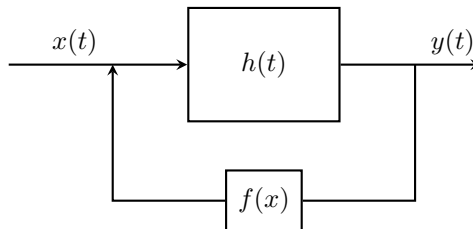
- **Linearity:** If the input is multiplied by a constant, the output is multiplied by the same constant.
- **Time-invariance:** If the input is shifted in time, the output is shifted by the same amount of time. In other words, the time at which the input is provided does not affect the system's output.

A simple diagram of a system is:



where $x(t)$ is the input, $h(t)$ is the system, and $y(t)$ is the output. The system is often considered a "black box" meaning we can only infer the properties of the system by observing the input and output.

A system can also have a feedback from the output to the input, represented by $f(x)$:



System theory involving feedback is called **control theory**.

10 Perturbation Theory

The notation from system theory carries over:

- $x(t)$ is the input
- $y(t)$ is the output
- $h(t)$ is the system

10.1 Dirac-Delta Function

$$\delta(t) \equiv \begin{cases} \infty & t = 0 \\ 0 & t \neq 0 \end{cases}$$
$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

The Dirac-Delta function is continuous, which is useful in analog applications.

10.2 Kronecker Delta Function

$$\delta_{ij} \equiv \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

The Kronecker Delta function is discrete, which is useful in digital applications such as digital signal processing (DSP).

10.3 Step Function

$$u(t) \equiv \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$
$$\frac{d}{dt}u(t) = \delta(t)$$

Also known as the Heaviside step function.

10.4 Rectangle Function

$\text{rect}\left(\frac{t-t_0}{a}\right) \equiv$ (a rectangle centered at $x = t_0$ with width a)

$\text{rect}\left(\frac{t-X}{Y}\right) \equiv u(t-X+Y/2) - u(t-X-Y/2)$

10.5 Sinc Function

$$\text{sinc}(x) \equiv \frac{\sin x}{x}$$

10.6 Convolutions

To get the output from the input:

$$y(t) = x(t) * h(t)$$

where $*$ is the convolution operator, defined as:

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

Convolutions can be informally thought as "smearing" two things together. Convolutions can be computed mathematically or graphically. In this class, we will only compute convolutions graphically.

Properties:

- When $x(t) = \delta(t)$, $y(t) = h(t)$.
- After many convolutions, the output will approach a Gaussian distribution.
- A delay circuit is when a function is shifted without distortion.

11 Fourier Transform

$$\mathcal{F}\{x(t)\} \equiv X(\omega) \equiv \int_{-\infty}^{\infty} x(t)e^{-i\omega t}dt$$

The **Fourier transform** $\mathcal{F}$ maps a signal from the time domain to the frequency (ω) domain. It can also be used on spatial data. A key advantage of the Fourier transform is data compression: frequency values often take up much less storage space than a full representation of the signal in the time domain.

The convention is to use a lowercase function like $x(t)$ in the time domain and an uppercase function like $X(\omega)$ in the frequency domain.

The integral usually yields a real and an imaginary component. The imaginary component contains phase information. However, we are usually only interested in the real component since that yields frequency information.

11.1 Discrete Fourier Transform

$$\mathcal{F}\{x(n)\} \equiv X(k) \equiv \sum_{n=0}^{N-1} x(n) e^{-i \frac{2\pi kn}{N}}$$

where N is the number of data points.

The **discrete Fourier transform (DFT)** is suitable for computers. However, in practice, software uses the FFT.

11.2 Fast Fourier Transform

If $N = 2^q$ for some integer q , then the DFT can be computed much faster using the **fast Fourier transform (FFT)**. Whereas the regular Fourier transform and DFT take $O(N^2)$ time to compute, the FFT takes $O(N \log N)$ time to compute. Any number of data points can be brought to the next highest power of 2 by padding the ends with zeros (a technique called **zero-padding**).

11.3 Fourier Transform Table

There are many Fourier transform pairs, but these are the ones we will need in this class.

$x(t)$	$X(\omega)$
1	$2\pi\delta(\omega)$
$\delta(t)$	1
$\text{rect}(t/T)$	$T \text{sinc}(\frac{T}{2\pi}\omega)$
$\text{sinc}(\omega t)$	$\frac{1}{\omega_0} \text{rect}(\frac{\omega}{2\pi\omega_0})$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$-i\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$

11.4 Convolution Theorem

$$x(t) * h(t) = \mathcal{F}^{-1} \{ \mathcal{F} \{x(t)\} \mathcal{F} \{h(t)\} \}$$

Notice that multiplication in the time domain is equivalent to convolution in the frequency domain, and convolution in the time domain is equivalent to multiplication in the frequency domain.

This theorem is useful in because convolutions are much slower to compute than the Fourier transform.

11.5 PSD

[work in progress - this topic will be covered later in the class]

$$\text{PSD}(\omega) = |X(\omega)|^2 = (\text{real})^2$$

11.6 Windowing Functions

The Fourier transform inherently messes with the spectral content of a signal. To mitigate this, we can use windowing functions.

$$\mathcal{F} \{x(t)\} = \mathcal{F} \{x(t)w(t)\} = X(\omega) * W(\omega)$$

where $w(t)$ is the windowing function.

The perfect window needs an infinite number of data points, which is impossible to achieve in practice. Instead, we want to minimize the **width** of the **main lobe** in a dB-vs-frequency plot of the window function.

11.6.1 Hanning Window

$$W(n) = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi n}{N-1} \right) \right]$$

PSD has multiple full-forms:

- Power Spectrum
- Power Spectrum Distribution
- Power Spectral Distribution
- Power Spectral Density

Each are technically defined differently, but are often used interchangeably. We will stick to PSD to avoid any confusion.

12 Random Facts

These random facts may appear on exams to separate an "A" student from the "B" student.

- Turbulence measurements often follow the Rayleigh distribution.
- The mean for the Cauchy distribution is undefined.